

Evaluating and Equivalent Expressions

Teacher's Edition

Read This First

This period is Math Nation 3.6 and 3.7 taught as one lesson (scope and sequence, ruling 14: paired book lessons run as one period). 3.6 is 'evaluate with several laws'; 3.7 is 'rewrite to a common base and decide equivalence'. Notes I-II carry 3.6, Notes III-IV carry 3.7, and the round mixes the two.

Everything on the book's pages for both lessons checks. The one defect in the surrounding material is on their homework key for 3.6 (nothing wrong) — no repairs were needed here; the audit list below records what was verified.

The book's 3.7 needs calculators for its exploration ($2^{14}/2^4$, $4^7 \cdot 4^{-2}$, $16 \cdot 4 \cdot 4^2$ all equal 1024). Here the equality is shown by rewriting to base 2, which is the point; the calculator is a check.

Standard and Targets

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Clarification Integer exponents, rational bases, procedural fluency — this is the fluency day. The state guide's phrase is 'with and without a calculator'; the round is the without.

Boundary Bases that are rewritten are powers of 2, 3 or 5 (the Unit 2 lists). No prime factorization of arbitrary numbers.

Learning target I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

Essential question How do you determine whether two expressions are equivalent when they do not have the same base?

Building on 3.01–3.05 — the seven laws with integer exponents and rational bases; Unit 2 — perfect squares and cubes ($8 = 2^3$, $9 = 3^2$, $64 = 2^6 = 4^3 = 8^2$).

Working toward Thread A (the next two days) — the same laws on variable bases, MA.8.AR.1.1; Unit 4 — operations in scientific notation.

Where this sits in the course Third and fourth MA.8.NSO.1.3 lessons, merged. The two Thread A days that follow do this same work with variable bases (MA.8.AR.1.1).

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **16 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-7	14 min
Example 1	8-9	3 min
Your Turn	10-11	3 min
Example 2	12-13	3 min
Your Turn	14-15	3 min
Whiteboards	16-33	16 min
IXL	34	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything and are how students confirm $8^5 = 2^{15} = 32,768$ once.

Where the lesson lives Notes III — a base can be rewritten as a power of a smaller base, and that is how two expressions with different bases get compared. Whiteboard 9 (Xavier) is the test.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

common base — one base both expressions can be written with: 8^5 and 2^9 share the base 2, because $8 = 2^3$

equivalent — same value; the laws of exponents turn one form into the other

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$ $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$

retrieves: yesterday — 3.05 — flip the base

2. Find the value: $\frac{1}{5^{-2}}$ **25**

retrieves: earlier this week — 3.04 — the negative exponent law, second form

3. Find the value: $\frac{10^6}{10^4}$ **100**

retrieves: earlier this unit — 3.02 — quotient of powers

4. Write 64 as a power of 2, of 4, and of 8. 2^6 , 4^3 , 8^2

retrieves: last unit — the prerequisite this lesson needs — Unit 2 — the perfect-square and perfect-cube lists (2.01, 2.03)

Slide-by-Slide Teaching Notes

Slide 1. Evaluating and Equivalent Expressions 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the prerequisite for the second half of today: one number, three bases. A board that stops at 8^2 has not yet seen that 8 is itself a power of 2 — that is Notes III.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes 3 min

- Melanie applies the product law first ($-2 + 5 = 3$), then the quotient law. Adonis rewrites 5^{-2} as $1/25$ first and does arithmetic. Both reach 25. The teaching point from 3.03 again: apply the laws BEFORE evaluating when you can — the numbers stay small.

Slide 5. Notes — Copy the worked line. 4 min

- Power of a power ($2 \cdot -3 = -6$), then quotient ($-6 - (-5) = -1$), then the negative exponent law to finish. Then the error-analysis pair from the book: the first student multiplied exponents where the product law adds; the second flipped a fraction without flipping the exponent. Both are on the board as 'find the step'.

Slide 6. Notes 4 min

- 8 is 2^3 , so 8^5 is $(2^3)^5 = 2^{15}$. 9 is 3^2 , so $9^4 = 3^8$. And $2^3 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$. The table of squares, cubes and fourth powers from Unit 2 is the tool; the move is power of a power. Ask: which bases can be rewritten as powers of 2? (4, 8, 16, 32, 64.) Of 3? (9, 27, 81.)

Slide 7. Notes 3 min

- Xavier says both expressions rewrite to 2^{-7} . Rewrite each: $(1/4)^2 = (2^{-2})^2 = 2^{-4}$, times 2^{-3} is 2^{-7} . $(1/2)^4 = 2^{-4}$, times $4^{-1} = 2^{-2}$ is 2^{-6} . Not equivalent — same base makes the comparison honest. This is the shape of whiteboard 9.

Slide 8. Example 1 — Evaluate. 1 min

- SPOKEN ONLY. What does the negative exponent do to $3/4$? Is the 3 outside part of the base? (No — it is a separate factor.)

Slide 9. Example 1 — Worked. 2 min

- Flip, square, then multiply by 3. A student who writes $(9/4)^{-2} \cdot 3$ or $(3/4)^{-6}$ pulled the 3 into the base. $16/3$ is right; $5\frac{1}{3}$ is the same value.

Slide 10. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- Catches a student who folds the outside 2 into the base (board shows $(4/5)^{-2}$ or $4/25 \cdot 2$) and one who never flipped ($4/25 \cdot 2 = 8/25$). $25/2$ is right.

Slide 11. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 12. Example 2 — Equivalent or not? 1 min

- SPOKEN ONLY. The two expressions have bases 3 and 6 on the left, and 3, 2 and 2 on the right. What base can 6 be broken into? Then both sides can be written with 3s and 2s only.

Slide 13. Example 2 — Worked. 2 min

- Left: $6^3 = (3 \cdot 2)^3 = 3^3 \cdot 2^3$, so $3^5 \cdot 2^3$. Right: $(3 \cdot 2)^5 = 3^5 \cdot 2^5$, over 2^2 leaves $3^5 \cdot 2^3$. Identical, so equivalent; the value is $243 \cdot 8 = 1944$ if anyone wants it.

Slide 14. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- Catches a student who cannot split 10 into $2 \cdot 5$ (board stops at 10^3) and one who subtracts the wrong exponent on the right. Both are $2^5 \cdot 5^3$; yes.

Slide 15. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 16. Whiteboards · Question 1 of 9 16 min for the round

- Zero exponent then quotient. A board with 343 dropped the sign; 'undefined' read $(-7)^0$ as 0.

Slide 17. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 18. Whiteboards · Question 2 of 9

- The book's Danny/Reina expression. $-3 - (-2) = -1$, then flip. A board with $3/4$ forgot to flip; $(3/4)^{-5}$ subtracted the wrong way.

Slide 19. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 20. Whiteboards · Question 3 of 9

- The book's choose-your-own-adventure expression: four laws. Everything becomes a power of 2: $-3 + 4 - 3 = -2$. A board with 4 forgot the final flip; a board with 2 miscounted the denominator as 2^2 .

Slide 21. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 22. Whiteboards · Question 4 of 9

- Notes III, plain. A board with 2^8 added $3 + 5$; a board with 2^5 replaced 8 by 2.

Slide 23. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 24. Whiteboards · Question 5 of 9

- Diagnostic. $9 = 3^2$, so $(3^2)^4 = 3^8 = 6561 = 9^4$. B is the common board.

Slide 25. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 26. Whiteboards · Question 6 of 9

- Diagnostic; this is Xavier's first expression. $(1/4)^2 = (2^{-2})^2 = 2^{-4}$, times 2^{-3} .

Slide 27. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 28. Whiteboards · Question 7 of 9

- Common base by rewriting one factor. A board with 8^6 or 8^3 multiplied the bases.

Slide 29. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 30. Whiteboards · Question 8 of 9

- Product of powers, then common base. 1024 either way. A board with 2^5 forgot to double the exponent when 4 became 2^2 .

Slide 31. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 32. Whiteboards · Question 9 of 9

- Written. Full credit: both rewrites shown and the sentence. A student who agrees with Xavier usually wrote 4^{-1} as 2^{-1} — the same slip as option B on question 6.

Slide 33. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 34. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones.

Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$	-343	343 — sign dropped on an odd power; 'undefined' — $(-7)^0$ read as 0 [B1G-M 8.NSO.1.3 zero-exponent misconception]
2	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$	$\frac{4}{3}$	$3/4$ — $(3/4)^{-1}$ left unflipped [negative exponent law]; $(3/4)^{-5}$ or $(4/3)^5$ — subtracted $-3 - 2$ [integer subtraction of a negative]
3	$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$	$\frac{1}{4}$	4 — 2^{-2} left unflipped [negative exponent law]; $1/2$ — the denominator $2 \cdot 2^2$ counted as 2^2 [product of powers]
4	8^5	2^{15}	2^8 — added $3 + 5$ in the power-of-a-power step [confused with product of powers]; 2^5 — replaced the base 8 by 2 without the exponent 3
5	Which expression is equivalent?	A — 3^8	(B) added $2 + 4$ instead of multiplying [power of a power confused with product of powers]; (C) replaced 9 by 3 and kept the exponent [the 3^2 was not written]; (D) squared the exponent, $4^2 = 16$ [the outer exponent applied to the exponent]
6	Which expression is equivalent?	A — 2^{-7}	(B) wrote $1/4$ as 2^{-1} instead of 2^{-2} [the base 4 is 2^2 , not 2]; (C) dropped both minus signs [negative exponent law: a negative exponent does not disappear]; (D) multiplied $-4 \cdot 3$ [confused with power of a power]
7	$2^3 \cdot 4^3$	2^9	8^3 or 8^6 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; $2^6 - 4^3$ rewritten but the 2^3 dropped
8	$4^7 \cdot 4^{-2}$	2^{10}	$2^5 - 4$ replaced by 2 without the exponent 2; 4^9 or 2^{18} — subtracted instead of adding a negative
9	Xavier says $(1/4)^2 \cdot 2^{-3}$ and $(1/2)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree?	No — the second is 2^{-6}, not 2^{-7}	'Yes' — 4^{-1} rewritten as 2^{-1} [the base 4 is 2^2]

Question Bank — variation and answers

Variation. Questions 1–5 are one-expression evaluations that each add a feature: a decimal base, a negative fraction base with a negative exponent, a power of a power to a negative, a fraction that simplifies to a whole base, and a product raised to -1 . 6–10 name the laws as well as the value. The pattern breaks at 11, where the task turns around (write with a GIVEN base), and at 12, where a claim about equivalence is tested by rewriting. The Additional sheet mirrors every position and its Paul item is the non-equivalent case where the bank's Marquise item is equivalent.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

1. $\frac{0.25^3}{0.25^3} = 0.25^0 = 1$
2. $(-3)^3 \cdot \left(\frac{-3}{5}\right)^4 = \frac{(-3)^7}{5^4} = -\frac{2187}{625}$
3. $3^{-9} = \frac{1}{3^9} = \frac{1}{19,683}$
4. $8^{-2} = \frac{1}{64}$ (or $\left(\frac{6}{48}\right)^2 = \frac{36}{2304} = \frac{1}{64}$)
5. $3^{-4} \cdot 2^4 = \frac{2^4}{3^4} = \frac{16}{81}$
6. **36 — product of powers, quotient of powers**
7. **-9 — zero exponent, quotient of powers**
8. **25 · 4 = 100 — negative exponent**
9. $(-2.5)^3 = -15.625$ — **zero exponent, quotient of powers**
10. $\left(\frac{5}{7}\right)^4 = \frac{625}{2401}$ — **negative exponent, power of a quotient**
11. **(a) 2^9 (b) 5^6 (c) 8^4 (d) 2^8**
12. **(a) $2^{-9} \cdot 2^{-3} = 2^{-12}$ (b) $2^{-10} \cdot 2^{-2} = 2^{-12}$ (c) Yes — both are 2^{-12} .**
13. **A A**
14. **(a) $n = 4$ (b) $n = -3$**

Question Bank — Additional

1. $4^2 \cdot 4^4 = 4^6 = 4,096$
2. $\frac{(-5)^{12}}{(-5)^{-1}} = (-5)^{13}$ (**= -1,220,703,125**)
3. $\left(\left(\frac{2}{3}\right)^2\right)^3 = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$
4. $\left(\frac{1}{3}\right)^9 = \frac{1}{19,683}$
5. $(2^{10})^{-2} = 2^{-20} = \left(\frac{1}{2}\right)^{20}$
6. $\frac{3^5 \cdot 1}{3^{-6}} = 3^{11} = 177,147$
7. $\frac{2.3^{-4}}{2.3^{-4}} = 1$

8. $\frac{(\frac{1}{4})^{-6}}{(\frac{1}{4})^{-4}} = (\frac{1}{4})^{-2} = 16$
9. $\frac{4^{-3} \cdot 4^4}{4^3} = 4^{-2} = \frac{1}{16}$
10. (a) 3^{12} (b) 2^{18} (c) 4^{11} (d) 3^{-2}
11. (a) $2^{-6} \cdot 2^{-2} = 2^{-8}$ (b) $2^{-8} \cdot 2^{-2} = 2^{-10}$ (c) **No — only the first is 2^{-8} ; the second is 2^{-10} .**
12. **A A**
13. (a) **$n = 4$** (b) **$n = -5$**

What the audit against Math Nation found

- Math Nation 3.6 pages verified: 25, 1/6, 25, 16/3, -343, 4/3, 1/4, 3/2, and the two error-analysis rows (correct values 1/125 and 64/27).
- Math Nation 3.7 pages verified: 1/16, 25/4, 1024 three ways, 2^{15} , 3^8 , 2^9 , $3^5 \cdot 2^3$, 2^{-7} vs 2^{-6} , the partner table (3^7 , $2^2 \cdot 7$, 4^3 , $3^3/2^2$, $2 \cdot 5^2$).
- Practice L6 and L7 keys check; Homework L6 and L7 keys check.
- The book's 3.6.1 warm-up (place digits in boxes to make a true statement) is sound; replaced by spaced retrieval because it previews the lesson instead of retrieving.
- The book's 3.7.5 partner table is in the round as questions 7-8 and in the bank as question 11.

What changed from the book

- 3.6.1 and 3.7.1 warm-ups → one retrieval warm-up.
- 3.6.2 (Melanie/Adonis) and 3.6.4 (Danny/Reina) → Notes I and whiteboard 2.
- 3.6.5 Choose Your Own Adventure and 3.6.6 Error Analysis → whiteboard 3 and Notes II.
- 3.7.2-3.7.4 → Notes III-IV and Example 2.
- 3.7.5 partner collaboration → whiteboard 7-8 and bank question 11.
- Both cool-downs → whiteboard 9, written.